Modern Computer Algebra

Computer algebra systems are now ubiquitous in all areas of science and engineering. This highly successful textbook, widely regarded as the "bible of computer algebra", gives a thorough introduction to the algorithmic basis of the mathematical engine in computer algebra systems. Designed to accompany one- or two-semester courses for advanced undergraduate or graduate students in computer science or mathematics, its comprehensiveness and reliability has also made it an essential reference for professionals in the area.

Special features include: detailed study of algorithms including time analysis; implementation reports on several topics; complete proofs of the mathematical underpinnings; and a wide variety of applications (among others, in chemistry, coding theory, cryptography, computational logic, and the design of calendars and musical scales). A great deal of historical information and illustration enlivens the text.

In this third edition, errors have been corrected and much of the Fast Euclidean Algorithm chapter has been renovated.

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For endless patience

To Mercedes Cappuccino

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